

Increasing extreme winds challenge offshore wind energy resilience

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Climate change is amplifying the intensity of extreme strong winds, threatening the development and resilience of offshore wind energy systems. The ability of wind turbines to endure such conditions is determined by the fifty-year return period wind speed (U_{50}), a key parameter for turbine design. However, the long-term trends and spatial variability of U_{50} across global oceans remain largely unexplored. Here, we utilize hourly ERA5 wind speed data at 100 meters above sea level from 1940 to 2023 to reveals a significant global increase in oceanic U_{50} of $0.016 \text{ m s}^{-1} \text{ yr}^{-1}$ ($p < 0.01$), with upward trends evident in 62.85% of coastal regions. Notably, over 40% of both commissioned and planned offshore wind farms in Asia and Europe have encountered wind speeds exceeding the design threshold of Class III turbines (37.5 m s^{-1}). More than half of these wind farms are situated in regions with increasing U_{50} trends, a pattern strongly associated with changes in tropical and extratropical cyclone activity under global warming. These findings underscore the critical need to adapt offshore wind energy infrastructure to withstand evolving wind extremes.

Offshore wind energy is a cornerstone of the global transition to renewable energy, with cumulative capacity expected to rise from 63 GW in 2022 to nearly 494 GW by 2030 (refs. 1,2). However, offshore wind farms operate in some of the most challenging environments, facing strong winds, waves, and ocean currents³. Many high-potential offshore wind regions, such as China, Japan, and the United States, are located in cyclone-prone areas, raising concerns about the resilience of offshore wind infrastructure⁴.

Recent studies reveal that extreme strong wind events over global oceans are intensifying under climate change. Strong tropical cyclones (TCs), particularly those with lifetime-maximum wind speeds above the 70th percentile of all recorded tropical cyclones, have shown a

significant upward trend in maximum wind speed⁵. In addition, extratropical cyclones (ETCs) are shifting poleward in both hemispheres⁶, with transient eddy kinetic energy showing a pronounced increase along the winter storm tracks in the Southern Hemisphere⁷ and a weakening along boreal mid-latitude summer storm tracks⁸. Under a warming climate, high-resolution global climate models project both an increase in the frequency of strong TCs (maximum wind speeds exceeding 50 m s^{-1}) and an approximate 3% increase in daily maximum wind speeds of ETCs per degree of warming in high-latitude regions^{9–11}. These changes are driven by enhanced latent heat flux¹², stronger upper-level winds¹³ and shifts in sea surface temperature (SST) patterns¹⁴. This strengthening wind environment

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threatens the safe operation of wind turbines throughout their multi-decade lifespans, potentially leading to damage, early retirements, and economic losses¹⁵. For instance, Typhoon Yagi (2024), the strongest autumn typhoon on record, caused the collapse of six wind turbines in Hainan, China and resulted in over US\$ 8 million in damages (see Supplementary Fig. 1 and Movie 1)¹⁶, highlighting the vulnerability of offshore wind farms to extreme winds.

The parameter used to characterize the maximum wind speed a turbine can endure during its operational lifespan is called the reference wind speed (U_{ref}), which is required to exceed the environmental fifty-year return period wind speed (U_{50}) at turbine hub height¹⁷. The International Electrotechnical Commission (IEC) categorizes turbines into Classes I, II, and III, with reference wind speeds of 50, 42.5, and 37.5 m s⁻¹, respectively¹⁸. Therefore, U_{50} is a key parameter for assessing extreme wind conditions that are significant for wind turbine design. Normally, U_{50} is derived from a Gumbel distribution, requiring long-term observational data for accurate estimation. However, sparse in-situ measurements¹⁹ and the low temporal resolution and inconsistencies of satellite-derived wind products^{20,21} limit the robustness of U_{50} assessment. Consequently, long-term and spatially homogeneous reanalysis datasets, such as ECMWF's ERA5 that provides historical wind speed data dating back to 1940, have been extensively used for U_{50} estimation^{17,22}. However, the pattern and magnitude of U_{50} changes remain poorly understood under the evolving and predominantly intensifying extreme wind environment, particularly in regions where offshore wind farms are operating or under development. Assessing whether existing and future turbines can endure changing wind extremes is essential for effective offshore wind energy planning and operations.

Here, we present a comprehensive analysis of the long-term trends and spatial variability of U_{50} across global oceans from 1940 to 2023. U_{50} is estimated by fitting a Gumbel distribution of annual

maximum wind speeds over multi-decadal window (> 20 years) to ensure robustness¹⁷, with trends analyzed by shifting the time window year by year (see *Methods* for details). Considering the systemic underestimation of extreme wind speeds in reanalysis datasets caused by grid smoothing²³, we apply a spectral correction method to recover the missing variability in the high-frequency spectrum of wind speed. Since SST is a key indicator of global warming, we examine its relationship with U_{50} changes to assess the influence of climate change. We exclude land areas from our analysis because U_{50} values over land are significantly lower than the IEC design thresholds, with a spatial average of just 22.57 m s⁻¹ (Supplementary Fig. 2). Our primary aim is to assess risks to offshore wind farms by comparing U_{50} values with IEC standards, focusing on both commissioned wind farms and development zones, especially in developing countries with abundant offshore wind potential but limited financial and technical capacity to harness this resource²⁴. By providing a detailed understanding of the evolving patterns of extreme wind over space and time, we aim to help developers and policymakers reduce investment risks and mitigate potential losses in regions experiencing significant increases in U_{50} .

Results

Extreme wind speed and trends over the ocean

U_{50} values in oceanic regions range from below 10 m s⁻¹ to over 50 m s⁻¹, with higher values predominantly occurring along tropical cyclone-prone tracks²⁵ and in high-latitude regions, especially the Northeastern Pacific, North Atlantic, and Southern Westerlies (Fig. 1a). Our analysis shows that 36.7% of global oceanic grid cells have U_{50} values exceeding the U_{ref} of a Class III wind turbine (37.5 m s⁻¹), while approximately 20.2% of grid cells exceed 42.5 m s⁻¹, the U_{ref} for Class II wind turbines (Fig. 1a). Only 0.8% of grid cells surpass 50 m s⁻¹, the U_{ref} associated with Class I wind turbines. For consistency, all U_{50} values reported in this

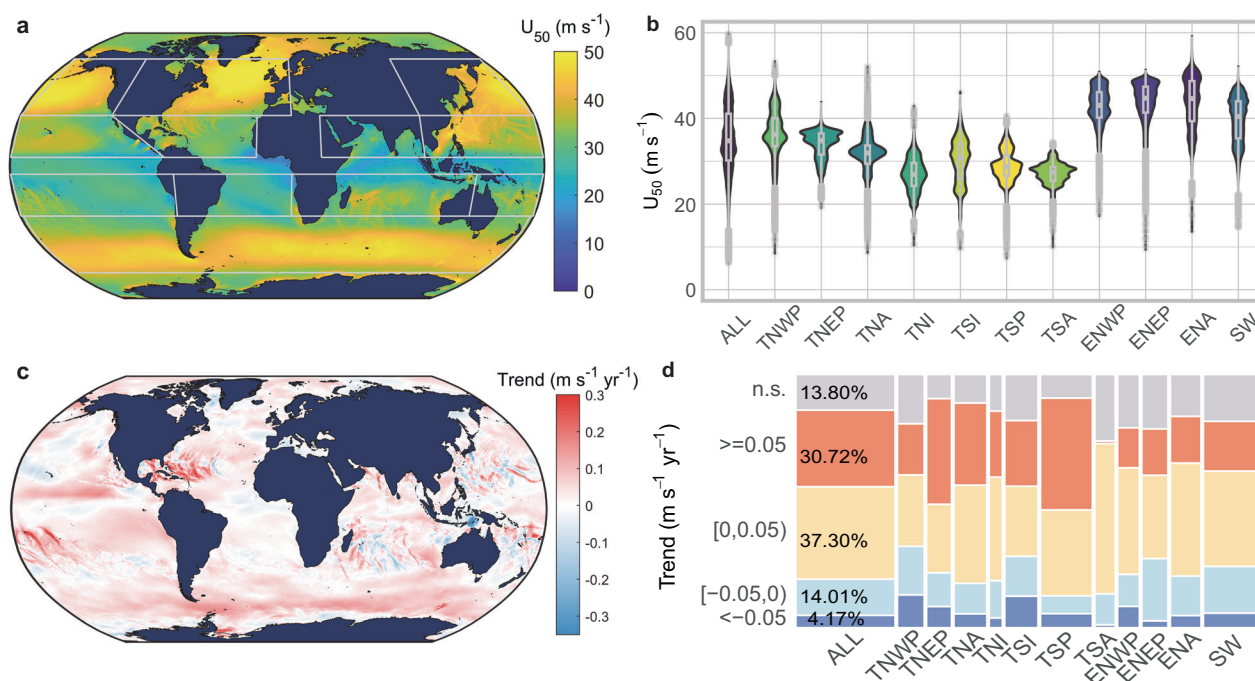


Fig. 1 | Spatial distribution and trend of the fifty-year return period wind speed (U_{50}). **a** The oceanic atlas of U_{50} . The land regions are covered with blue masks since we mainly focused on the oceanic areas. The 11 sub-basins used for regional analysis are outlined in white boxes, including the Tropical Northwest Pacific (TNWP), Tropical Northeast Pacific (TNEP), Tropical North Atlantic (TNA), Tropical South Indian Ocean (TSI), Tropical South Pacific (TSP), Tropical South Atlantic (TSA), Extratropical Northwest Pacific (ENWP), Extratropical Northeast Pacific

(ENEP), Extratropical North Atlantic (ENA), and South Westerlies (SW). **b** The distribution of U_{50} across all pixels and each sub-region. **c** The trend of U_{50} , with values estimated over a continuous 30-year period. The grey dots represent the significant linear trends. **d** The U_{50} trends in all pixels and each sub-region. The colors of the bars represent the trend values, and each bar width represents the area of the sub-region.

study are based on the complete 84-year time series, except for the analysis of U_{50} trends derived from the moving time window.

To analyze the spatial distribution of U_{50} , we classify ocean basins into tropical (5–30° N/S) and extratropical (30–60° N/S) regions, corresponding to the dominant meteorological regimes of TCs and ETCs, respectively. The southern extratropical Atlantic, Pacific and India Ocean are grouped into a unified Southern Westerlies region because they are well-connected and unobstructed by land, resulting in a total of 11 sub-basins (Fig. 1a). The U_{50} values in extratropical regions are substantially higher than those in tropical regions. A pronounced right-skewed distribution is observed in the Extratropical North Atlantic (ENA) and Southern Westerlies (SW) regions, indicating a higher frequency of extreme wind events (Fig. 1b). In tropical regions, U_{50} values are higher in the Pacific Ocean, with a broader distribution in the Tropical Northwest Pacific (TNWP) and a concentrated right-skewed distribution in the Tropical Northeast Pacific (TNEP). This spatial pattern aligns with the expansive cyclone-prone zone in the TNWP and the frequent occurrence of intense cyclones in the TNEP (Supplementary Fig. 3). These patterns are consistent with those observed in altimeter data²⁶ and reanalysis dataset from NCEP's Climate Forecast System Reanalysis²⁷.

A general upward trend in U_{50} is observed across most ocean regions. Globally, the average oceanic U_{50} has increased significantly at a rate of $0.016 \text{ m s}^{-1} \text{ yr}^{-1}$ over the 30-year window adopted in this research ($p < 0.01$, Fig. 1c). Trend analysis using continuous 20-year and 50-year windows yields similar rates of change ($0.014 \text{ m s}^{-1} \text{ yr}^{-1}$ and $0.013 \text{ m s}^{-1} \text{ yr}^{-1}$, respectively), suggesting that the choice of time window does not substantially affect the trend estimate (Supplementary Fig. 4). Spatially, 68.0% of oceanic grid cells exhibit a significant increasing trend, while only 14.01% show a decreasing trend (Fig. 1d). Regionally, the tropical regions, including the Northeast Pacific (TNEP), North Atlantic (TNA), North Indian Ocean (TNI), and South Pacific (TSP), exhibit a relatively high proportion of grid cells with substantial increases in U_{50} . In these regions, 42.77%, 33.03%, 26.35%, and 45.34% of grid cells exhibit an upward trend exceeding $0.05 \text{ m s}^{-1} \text{ yr}^{-1}$ (Fig. 1d).

Although this study primarily examines extreme wind speeds, we also assess trends in mean wind speed to identify regions that exhibit both enhanced wind energy potential and heightened extreme wind risks. The ERA5 dataset reveals a modest but widespread increase in mean wind speeds across most oceanic regions from 1940 to 2023, except for equatorial zones and the northern Indian Ocean (Supplementary Fig. 5a). Importantly, increases in mean wind speeds frequently coincide with intensifying extreme winds, particularly in the eastern tropical Pacific, southeastern Pacific, and Southern Westerlies. In addition, ERA5 provides hourly 3-second gust wind speed, which enables us to determine the origins of extreme wind—whether from sudden high-wind events or sustained strong winds. A significant widening of the difference is observed between gust winds and hourly mean winds (here we use the 90th percentile of the difference as an example), suggesting more frequent sudden high-wind events (Supplementary Fig. 5b). However, spatial discrepancies are observed in the north-central Pacific and northwest Atlantic when comparing ERA5-derived mean wind speed trends with those from altimeter observations²⁸ and the NCEI blended seawinds product²⁹. These differences may stem from temporal variations in data coverage, particularly in satellite-based products introduced after 1985.

Extreme wind speed and trend over the coastline

Given that offshore wind farms are primarily sited in coastal regions and that the farthest distance of commissioned wind farms is approximately 100 km from land³⁰, we specifically extract grid cells within 100 km of the coastline for analysis. In these coastal regions, around 16.49%, 11.25%, and 0.62% of grid cells exceed the reference wind speed thresholds for Class III, Class II, and Class I wind turbines,

respectively. The coasts of Canada, England, Japan, and southern Argentina have the highest proportion of U_{50} values exceeding the reference wind speed thresholds of wind turbine (Fig. 2a). Significant increases in extreme wind speeds, defined as trends larger than $0.05 \text{ m s}^{-1} \text{ yr}^{-1}$, are concentrated in tropical and subtropical coastal zones, including the Gulf of Mexico, the Caribbean, the southern and eastern United States, India, Southeast Asia, and northern Australia (Fig. 2b). Coastal regions experiencing increasing mean wind speed and concurrently strengthening extreme winds exhibit greater potential for offshore wind energy development, but also face heightened risks, such as the coasts of California, northern and eastern Europe, the Taiwan Strait, Africa, South America, and Australia (Fig. 2b, c). Conversely, concurrent declines in both mean and extreme wind speeds are detected in the Mediterranean Sea, East China, and coastal India, likely associated with the weakening and poleward shift of boreal mid-latitude storm tracks⁸.

The risk assessment for fully commissioned offshore wind farms

Changes in extreme oceanic winds pose increasing risks to offshore wind farms. To assess these risks, we categorize wind farms into three risk categories based on the U_{50} values of the grid cells in which they are located. These categories correspond to IEC reference wind speed thresholds: Type 1 (U_{50} exceeding 37.5 m s^{-1}), Type 2 (U_{50} exceeding 42.5 m s^{-1}), and Type 3 (U_{50} exceeding 50 m s^{-1}), as winds exceeding these thresholds endanger turbines in each respective category. Fully commissioned offshore wind farms are predominantly located in Europe and Asia, with 136 and 157 wind farms, respectively, and with total installed capacities of 28.86 GW and 25.24 GW³¹. Our analysis focuses on U_{50} values and their trends at these wind farm locations. Notably, given the large spatial coverage of $0.25^\circ \times 0.25^\circ$ grid cells, multiple wind farms may be situated within the same or neighboring grid cells.

In Europe, U_{50} values across commissioned wind farms are closely clustered around the mean of 40 m s^{-1} . The majority of these wind farms (74.26%) are located in regions exhibiting increasing U_{50} trends, particularly in key areas such as southeastern UK and the coastal waters of Germany, Denmark, Netherlands, Belgium, and Sweden (Fig. 3a). Specifically, 48 wind farms (total capacity of 9.68 GW) are exposed to Type 1 risk, while 61 wind farms (total capacity of 18.14 GW) face Type 2 risk. Among these high-risk farms, 75.23% are located in regions where U_{50} is trending upward. A temporal analysis shows a sharp increase in U_{50} around 1975, followed by a temporary decline during 2005–2015, and a renewed upward trend in the past decade (Supplementary Fig. 6a).

In contrast, commissioned wind farms in Asia exhibit greater spatial variability in U_{50} , with values ranging from below 15 m s^{-1} to above 40 m s^{-1} (Fig. 3b). Over half of these sites show decreasing U_{50} trends, with spatial variance nearly ten times greater than that observed in Europe. Fewer wind farms are currently at risk in Asia, with eight sites falling under Type 1 risk and two wind farms near the Taiwan Strait exposed to both Type 2 and Type 3 risks (total capacity of 1.45 GW). Among these risk-exposed Asian wind farms, U_{50} generally shows an upward trend, though it was briefly interrupted around 1985. However, two wind farms located in South China deviate from this pattern, with trends remaining persistently flat or negative (Supplementary Fig. 6b).

The risk assessment for offshore wind farm development zones

The U_{50} values and their trends exhibit broadly similar patterns across offshore wind farm development zones. In Europe, U_{50} trends are predominantly characterized by significant increases (68.03% of grid cells), mainly concentrated in the North Sea, the Baltic Sea, and along the west coast of England. Conversely, declining U_{50} values are primarily observed along the coastlines of Spain and Italy (Fig. 4a). Among the planned wind farms, 208 are exposed to Type 1 risk, while

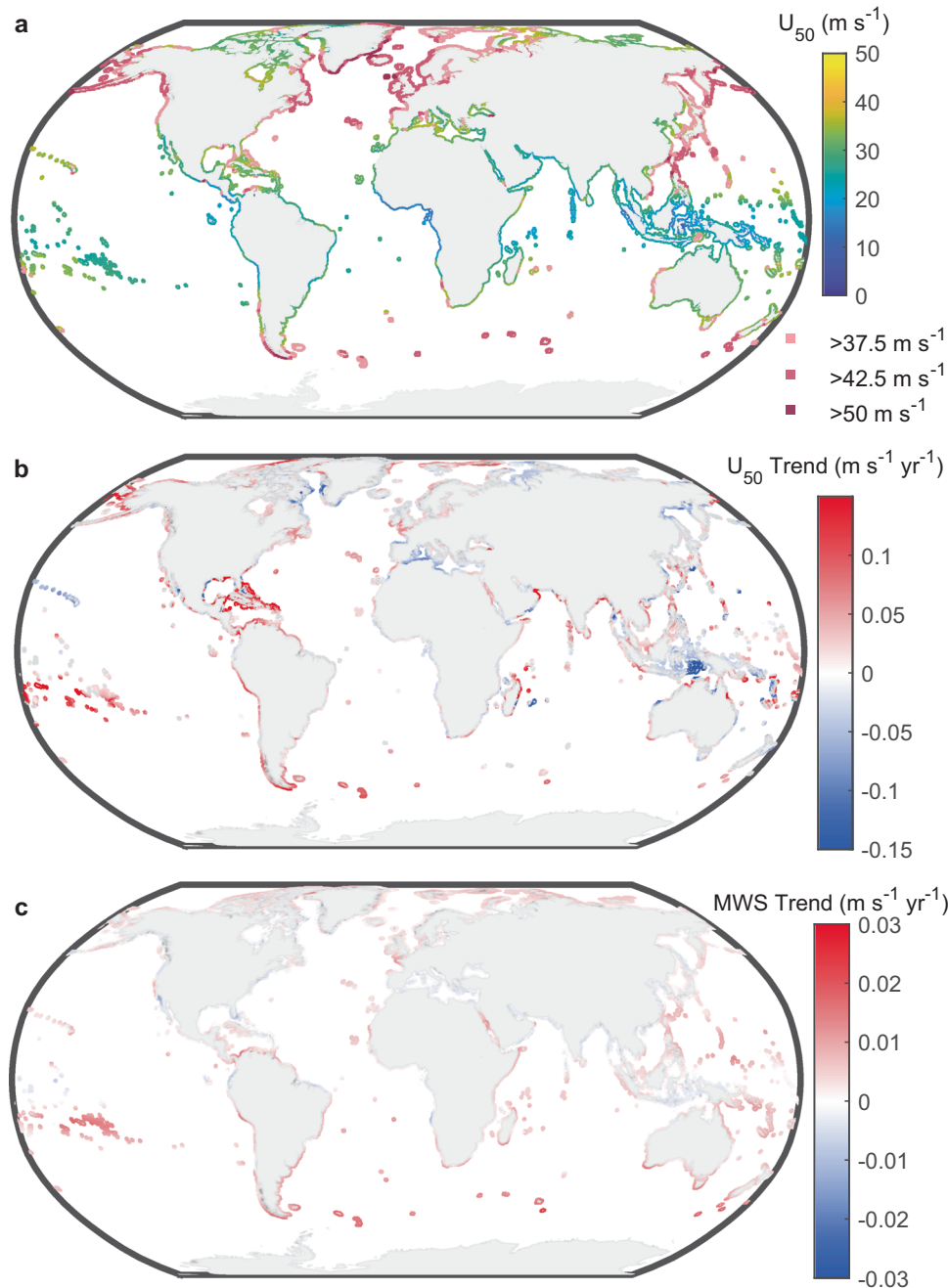


Fig. 2 | The fifty-year return period wind speed (U_{50}) and its trend in coastal regions. a Coastal U_{50} . The pastel pink, soft pink, and deep pink represent U_{50} exceeding 37.5 m s^{-1} , 42.5 m s^{-1} , and 50 m s^{-1} , respectively. **b** The trend of U_{50} in

coastal regions. **c** The trend of mean wind speed (MWS) in coastal regions. Since the farthest offshore distance of commissioned wind turbines is 100 km, the coastal regions analyzed here are defined as areas within 100 km offshore.

292 face Type 2 risk, corresponding to a combined maximum planned capacity of 403.00 GW. Of these, 76.80% are located in areas with rising U_{50} trends. The temporal evolution of U_{50} across these wind farms mirrors the pattern observed in commissioned wind farms, characterized by a general upward trend interrupted by a temporary decline during 2005–2015 (Supplementary Fig. 7a).

In Asia, 61.23% of planned wind farms are located in regions with increasing U_{50} trends, particularly concentrated in South Korea, Vietnam, the eastern and southeastern coasts of China (Fig. 4b). Although the mean U_{50} value in planned wind farms across Asia is approximately 4 m s^{-1} lower than that in Europe, several projects in Asia are exposed to higher risk categories. Specifically, 72 wind farms (total maximum planned capacity of 56.89 GW) face Type 1 risk, 46 wind farms (total

maximum planned capacity of 42.11 GW) face Type 2 risk, and 69 wind farms (total maximum planned capacity of 40.52 GW) face Type 3 risk (Fig. 4b). Over 67% of these projects are located in regions where U_{50} is increasing. The temporal trends follow a similar pattern to those in Europe, showing a sustained decline prior to the 1990s followed by a rebound after the 1990s (Supplementary Fig. 7b).

There are a total of 172 planned wind farms in North America, South America, and Australia. They are primarily located along the eastern and western coasts of the United States, the northeastern and southeastern coasts of Brazil, and the southwestern and southeastern coasts of Australia (see Supplementary Fig. 8)³¹. The increasing trends of U_{50} are most pronounced in the northeastern United States and in southeastern and southwestern Australia, while most other sites show

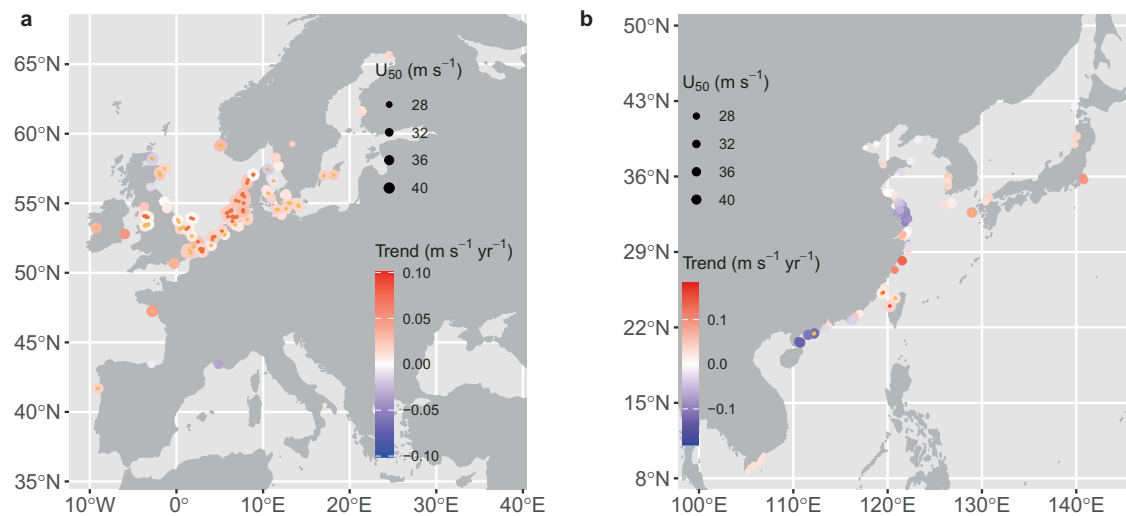


Fig. 3 | The fifty-year return period wind speed (U_{50}) and its trend in fully commissioned wind farms. a U_{50} and trend for Europe. **b** U_{50} and its trend for Asia. The solid-colored dots represent the locations of wind farms in Asia and Europe. The colors of the dots represent the magnitude of the trend, and the size of the dots

represents the magnitude of U_{50} at each wind farm. Additionally, smaller yellow, orange, and red dots mark wind farms that meet Type 1, Type 2, or Type 3 risk criteria, respectively.

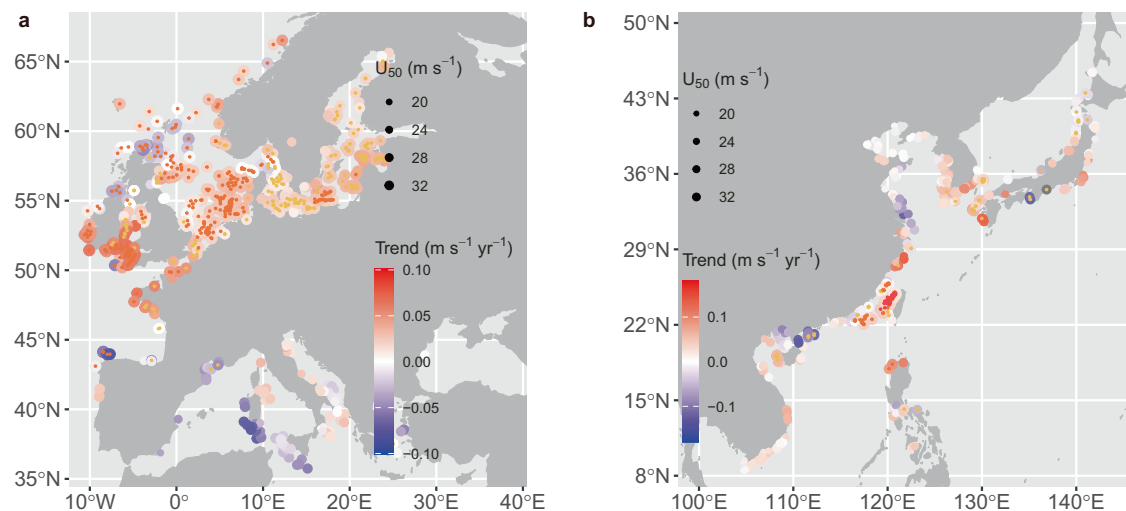


Fig. 4 | The fifty-year return period wind speed (U_{50}) and its trend in offshore development zones. a U_{50} and trend for Europe. **b** U_{50} and trend for Asia. The solid-colored dots represent the locations of wind farms in Asia and Europe. The colors of the dots represent the magnitude of the trend, and the size of the dots

represents the magnitude of U_{50} at each wind farm. Additionally, smaller yellow, orange, and red dots mark wind farms that meet Type 1, Type 2, or Type 3 risk criteria, respectively.

decreasing trends. Compared to Europe and Asia, these regions display smaller magnitudes of U_{50} trends. In the United States, more than half of the planned wind farms, with a total maximum planned capacity of 50.31 GW, are exposed to Type 1 and Type 2 risks. A substantial increase in U_{50} began to emerge around 2005 and has continued to date. The remaining sites fall within a relatively low-risk range for extreme wind (Supplementary Fig. 9).

The U_{50} details in developing countries

Given the critical role of developing countries in the global energy transition and in meeting the United Nations' Sustainable Development Goal 7 to ensure access to clean and affordable energy for all, we analyze extreme wind conditions in nine emerging offshore wind markets, including Brazil, Colombia, India, the Philippines, Romania, South Africa, Sri Lanka, Turkey, and Vietnam²⁴. These countries have set ambitious targets for offshore wind development, which will require substantial financial commitments. For example, Colombia is

planning 12 projects with a total capacity of 5 GW, scheduled for commissioning between 2032 and 2034 (refs. 32,33). India has issued tenders for offshore wind zones targeting 37 GW by 2030, with the transmission system for 10 GW estimated to cost around 0.28 lakh crore (approximately US\$ 3.36 billion)³¹. Vietnam aims for 6 GW by 2030, with foundation construction alone requiring investments of US\$ 40–100 million²⁴. Brazil has advanced regulatory decrees to facilitate the legal use of offshore areas for wind projects³². These developments underscore the need for accurate assessments of extreme wind risks to support efficient capital deployment in these countries.

Six of these nine countries, excluding Brazil, Turkey, and Romania, have over half of their coastal grid cells showing increasing U_{50} trends. In Turkey, U_{50} trends exhibit clear regional variations, with a significant downward trend observed near the Mediterranean Sea and upward trends near the Black Sea (Fig. 2b). In Sri Lanka, over 30% of grid cells show a trend greater than 0.05 m s⁻¹ yr⁻¹ (Fig. 5a). Colombia experiences different oceanic environments due to its exposure to

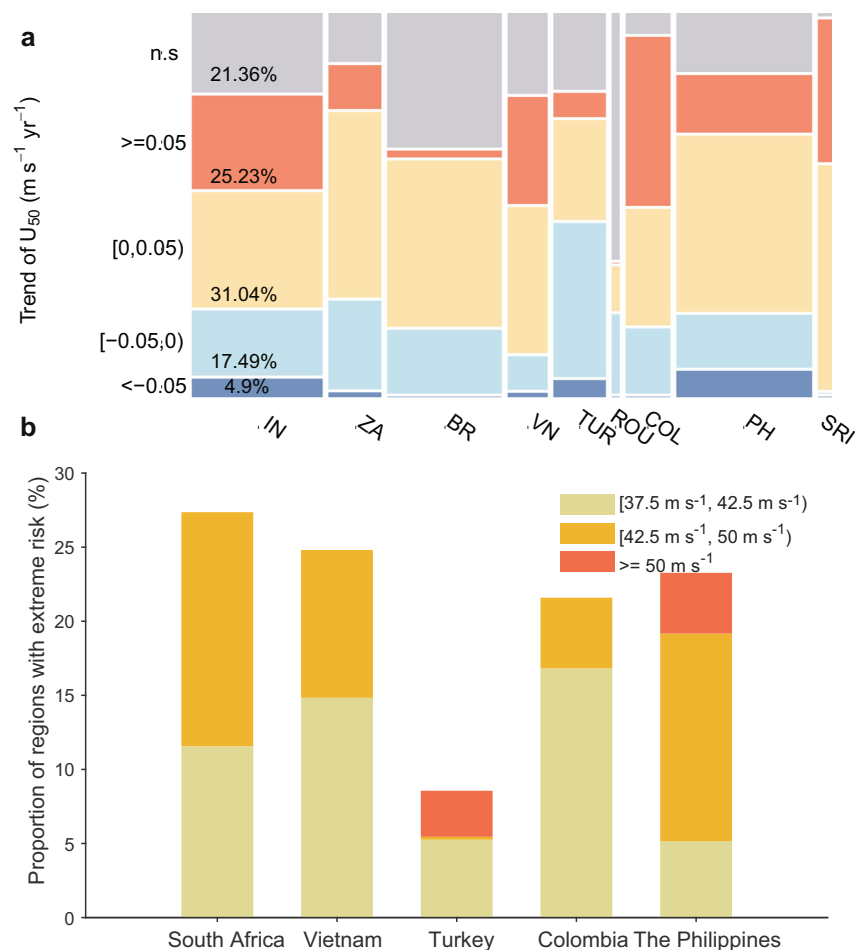


Fig. 5 | The extreme wind in nine developing countries coastal regions. a The trend of U_{50} in the coastal regions of nine developing countries. The abbreviations IN, ZA, BR, VN, TUR, ROU, COL, PH, and SRI represent India, South Africa, Brazil, Vietnam, Turkey, Romania, Colombia, the Philippines, and Sri Lanka, respectively.

The widths of the bars represent the relative amount of grid cells in each country. **b** The proportion of coastal areas where U_{50} exceeding reference wind speeds. The reference wind speeds for Class III, II, and I turbines are 37.5 m s^{-1} , 42.5 m s^{-1} , and 50.0 m s^{-1} , respectively.

both the Caribbean Sea and the southeastern Pacific Ocean, with extreme winds primarily concentrated along the Caribbean coast (Fig. 2b).

When compared to the reference wind speeds in IEC standards, five countries—South Africa, Vietnam, Turkey, Colombia and the Philippines—will face extreme risks (Fig. 5b). Notably, approximately 3% of coastal grid cells in both Turkey and the Philippines exceed U_{50} values of 50 m s^{-1} (Type 3 risk), with extreme wind conditions in Turkey mainly occurring in the Black Sea. Such high-wind environments pose additional challenges for floating offshore wind energy development, including increased maintenance difficulties and higher anchor costs³⁴.

Attribution of U_{50} trends

To explore potential drivers of the observed changes in U_{50} , we use the relationship between SST and annual maximum wind speed (AMW), which serves as a proxy for U_{50} at annual timescales. While SST is not a direct indicator of wind dynamics, it reflects low-frequency variability in the climate system and may influence storm intensity through thermodynamic and dynamic processes³⁵. Our analysis reveals a strong correlation ($r = 0.80$, $p < 0.05$) between spatially averaged SST and AMW at the global scale from 1940 to 2023 (Fig. 6a). Over this period, AMW increased by 1.05 m s^{-1} per degree of SST change, indicating a potential response of extreme winds to global warming (Fig. 6b).

The observed correlation between SST and AMW can be explained by the intensification of storm systems under a warming climate. This is because TCs and ETCs account for over 80% of extreme wind events^{11,36}. Our seasonal analysis indicates that most extreme wind events occur in winter months, suggesting a dominant contribution from ETCs (Supplementary Fig. 10)³⁷. Regions where extreme winds peak in summer or autumn align with known TC tracks, such as the North Indian Ocean, the tropical Pacific, and the tropical North Atlantic.

Observed changes in cyclone characteristics provide evidence of intensifying wind extremes. The spatial-temporal patterns of cyclone activities derived from best-track data (details refer to Supplementary Methods) are consistent with U_{50} changes. Although TC frequency peaked in the 1990s and 2000s, the 2010–2023 period recorded the highest lifetime-maximum wind speeds in most basins (Supplementary Fig. 11a, b). Similarly, accumulated cyclone energy (ACE), defined as $ACE = \sum_0^N \sum_0^{\tau} V_{\max}^2$ (where $V_{\max}$ is the maximum wind speed of cyclone, τ is cyclone lifetime, and N is the total number of cyclones)³⁸, also peaked during this period (Supplementary Fig. 11c). The spatial extent of regions experiencing Category 4 and 5 cyclones ($V_{\max}$ larger than 58 m s^{-1}) has expanded by 4.3% compared to the 1980–2001 baseline, particularly in the North Atlantic, consistent with the sharp increase in U_{50} observed in the Gulf Stream region (Supplementary Fig. 3). ETCs in both hemispheres are shifting poleward, with notable intensification over the northeastern Atlantic in the Northern Hemisphere and near

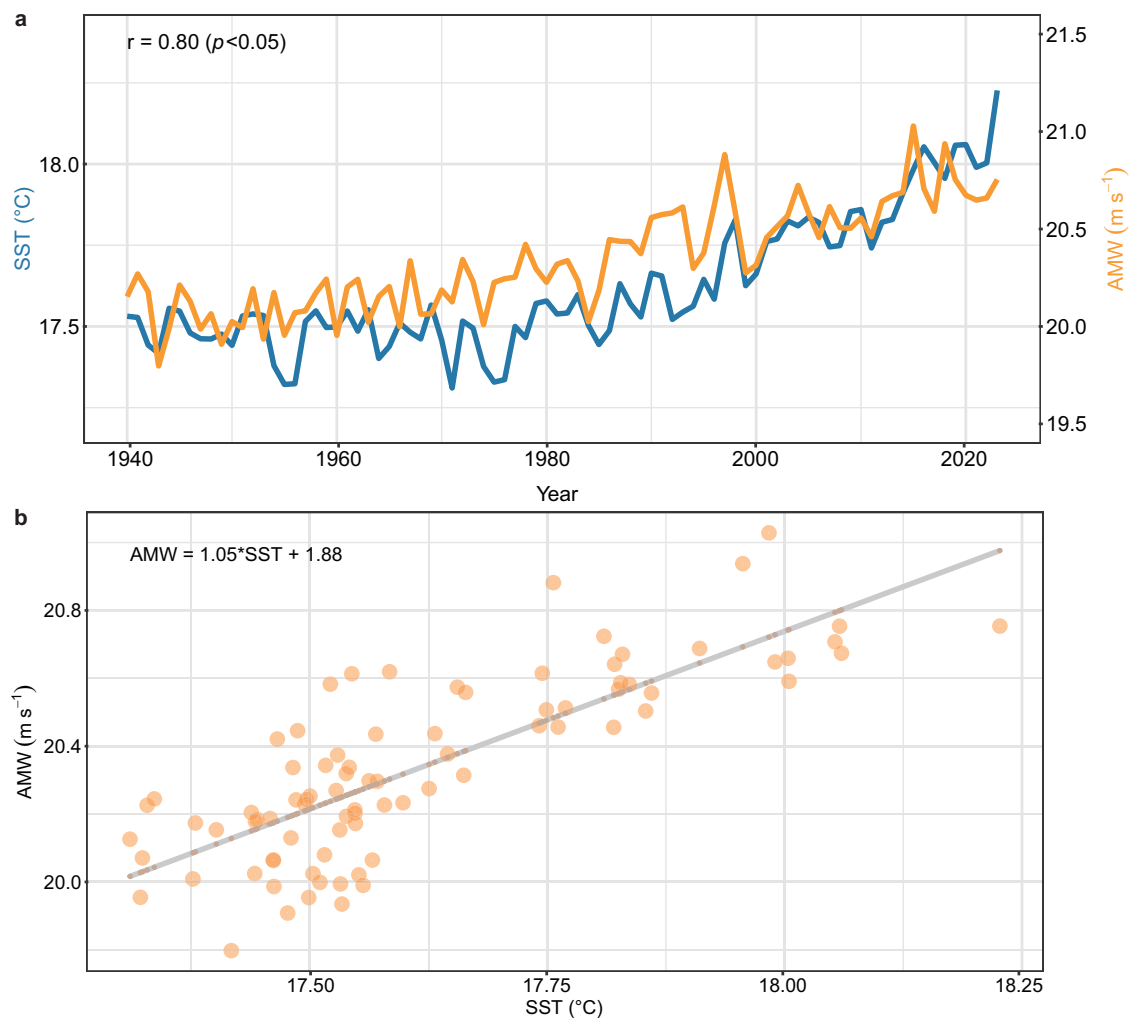


Fig. 6 | Statistical relationship between sea surface temperature (SST) and annual maximum wind speed (AMW). **a** Global time series of SST and AMW. **b** Scatterplot of SST versus AMW with linear regression. The grey line represents the fitted linear regression.

60°S in the Southern Hemisphere¹¹. Storm activities have weakened in midlatitude regions, consistent with the decreasing U_{50} trends around the Mediterranean Sea and China's Bohai Sea.

Several physical mechanisms may contribute to these changes, including enhanced latent heat flux from rising SST^{6,12} and increased positive anomalies of lower-tropospheric potential vorticity¹¹. For instance, TC intensification rates are reported to be three times higher for TCs generated under marine heatwave (MHW) environments compared to those under non-MHW conditions¹². Additionally, a warming climate intensifies upper-level winds and the meridional temperature gradient³⁹, further promoting the poleward shift of ETCs⁶. While future U_{50} projections remain uncertain and require comprehensive Earth System Model evaluations, the observed intensification of U_{50} is likely to continue in tandem with warming trends⁴⁰.

Discussion and implications

This study presents a comprehensive analysis of global extreme oceanic wind speeds (U_{50}), their long-term trends, and their potential influences from global warming. Using gridded hub-height wind speed data from ERA5 (1940–2023) corrected for underestimation biases via spectral correction, we find that U_{50} has significantly increased across 62.85% of global coastal regions over the past nine decades, with a global average trend of $0.016 \text{ m s}^{-1} \text{ yr}^{-1}$. Our results show that over 40%

of commissioned and planned wind farms in Asia and Europe are exposed to U_{50} levels exceeding IEC design thresholds, and more than 60% are located in regions experiencing rising U_{50} trends. The changes in U_{50} are closely linked to the evolving patterns of TCs and ETCs associated with global warming. Considering the continuous warming trend, regions with upward U_{50} trends require more attention to mitigate damages caused by intensifying extreme wind events.

This study has several limitations. The spectral correction (SC) method applied to AMW increases U_{50} by an average of 5.56 m s^{-1} (Supplementary Fig. 12b). However, the SC method tends to underestimate U_{50} in cyclone-prone regions, where original U_{50} values are retained and warrants further refinement. While the current assessment reveals many high-risk areas overall, wind risks in these regions remain conservative. For offshore wind farms located in cyclone-prone regions, a more detailed downscaling of extreme wind load assessments is recommended to improve design safety. Due to the limited availability of sufficient long-term observations, direct validation of current U_{50} results was not feasible; however, the SC method used here is consistent with that employed by Larsen et al.²³, which demonstrated strong agreement with in-situ measurements of AMW²³. This approach has been successfully applied in prior U_{50} assessments^{22,41,42}. Previous studies employing empirical formulas to correct reanalysis wind speed underestimation are limited by a relatively coarse temporal resolution of 1 hour and a lack of in-situ observations exceeding 25 m s^{-1} (ref. 29).

In contrast, the SC method effectively refines U_{50} to a 10-minute resolution, aligning with the IEC standard definition.

This study evaluates U_{50} at a hub height of 100 m. However, in practice, turbine heights vary and have steadily increased over the past two decades, averaging 104.03 m in 2020 (ref. 43), with the tallest turbine reaching 185 m in 2024. Using power-law extrapolations, we estimate U_{50} at hub heights of 80 m, 120 m, 150 m, and 200 m (Supplementary Fig. 13). Results show that U_{50} and associated risk increase consistently with height. For instance, the proportion of grid cells facing Type 1 risk rises from 32.48% at 80 m to 54.28% at 200 m, while Type 3 risk coverage expands tenfold when hub height increases from 100 m to 200 m. Although taller turbines improve energy yield and economic returns⁴⁴, they also heighten vulnerability to extreme wind events.

The U_{50} distribution and trends revealed in this study offer a reference for offshore wind design. However, turbine standards include more than just fifty-year return period wind speed. Other critical design criteria include extreme wind shear, extreme direction change, and extreme turbulence (IEC 61400-1). There are ongoing debates surrounding IEC standards, such as the load factors in IEC 61400-3 are based on north European climates and may not suit cyclone-prone regions^{45,46}. In addition, observed wind speed variability in coastal Denmark already exceeds the prescribed extreme turbulence model in the IEC 61400-1 Edition 3 standard⁴⁷, highlighting the need to localize design criteria based on regional wind behavior.

Extreme winds, especially from TCs and ETCs, are the leading cause of wind turbine failure, accounting for 54.9% of all reported failures⁴⁸. These failures can result in substantial economic losses, with damages ranging from US\$ 0.64 million to US\$ 6.4 million, depending on turbine type, equivalent to 10% of total investment⁴⁸. We recommend using higher design-load-class wind turbines in regions with intensified extreme winds. Additionally, resilience-informed planning is crucial to mitigate losses, limit failure propagation, and accelerate recovery, as the failure of one wind turbine can trigger a chain reaction of failures⁴⁹. In scenarios where damage from extreme winds is unavoidable, a “strong column-weak beam” design philosophy is suggested to minimize potential losses, with blades designed to fail first to protect more expensive components such as towers and nacelles⁵⁰.

In summary, this study highlights a widespread increase in extreme wind speeds across global oceans, with direct implications for offshore wind farm design, siting, and resilience planning. By quantifying U_{50} trends and identifying their spatial patterns, our findings offer actionable guidance for industry stakeholders, especially in vulnerable regions. As tropical and extratropical cyclones continue to evolve under global warming, proactive adaptation, including improved risk modeling, revised design standards, and robust turbine engineering, will be essential to safeguard offshore wind infrastructure and support its long-term expansion.

Methods

Long-term, gridded hub-height wind speed dataset

The ERA5 dataset is the fifth iteration of the atmospheric reanalysis dataset developed by the European Centre for Medium-Range Weather Forecasts (ECMWF), spanning the period from 1940 onward⁵¹. With a high spatial resolution of $0.25^\circ \times 0.25^\circ$, ERA5 provides hourly estimates of wind speed components alongside various meteorological variables. The assimilation process for ERA5 incorporates an extensive array of in-situ and remote-sensing observations. Specifically for wind speed, ERA5 integrates satellite retrievals from radiance data, in-situ measurements from radar detection, drifting buoys, ship stations, and aircraft observations⁵². Distinguished from its predecessors, ERA5 is produced using the latest version of the ECMWF

Integrated Forecast System (IFS), featuring enhanced model parameterizations and data assimilation methodologies. These improvements have been validated, demonstrating advancements in capturing wind patterns across global oceans^{52,53}.

The reliability of ERA5 wind speed data has been established through evaluations, particularly in offshore locations⁵⁴. It has widespread application in offshore wind resource estimation and extreme wind speed assessments. Examples include global offshore wind energy resource estimations by Soares et al.⁵⁵, assessments of extreme wind speeds in the U.S.⁵⁶, and on a global scale¹⁷. In this study, we extract hourly *u*- and *v*-components of wind speed data from ERA5, covering the period from 1940 to 2023 and at 100 m above the ground, near the wind turbine hub height. ERA5 is a dataset suitable for resolving U_{50} assessments because it provides data close to hub height with higher temporal resolution and sufficiently long records. To explore the differences in U_{50} at various hub heights, we apply the power law to extrapolate wind speed to different hub heights as an example (details refer to Supplementary Methods). In this research, we only consider and evaluate wind speed from 85°N to 85°S since ERA5 has been reported to have a negative bias in the *u*- and *v*-components at the poles⁵². For the coastal regions, a grid cell with a land area of less than 10% is defined as oceanic.

Additionally, ERA5 provides 3-second maximum-averaged wind speed for each hour, which is derived by the IFS from hourly averaged surface stress, surface friction, wind shear, and stability⁵⁷. Analyzing gust wind allows us to understand whether extreme wind originates from sustained strong winds or abrupt, intense gusts. Since ERA5 offers gust wind data at 10 m height, it is adjusted to 100 m using the power law (details refer to Supplementary Methods).

Sea surface temperature (SST)

SST is incorporated in this study to examine the relationship between changes in extreme wind and global warming. To maintain consistency, SST data were also obtained from ERA5, which assimilates the HadISST2 dataset before 2007 and OSTIA data thereafter, maintaining spatial and temporal resolutions consistent with wind speed data⁵⁷. ERA5 SST data show strong agreement with other datasets regarding the warming rate since the mid-1970s, with minor discrepancies observed between HadISST2 and the more recent HadSST4 over the preceding 10 years. The SST analysis is based on a global spatial average, weighted by latitude (details refer to Supplementary Methods).

Dataset of wind farm locations

Wind farm locations are integral to our risk assessment approach, wherein the U_{50} assessment is harmonized with offshore wind farm geographical data to provide a tailored risk evaluation for offshore wind farm development. We utilize information from 4C Offshore Ltd (<https://map.4coffshore.com/offshorewind/>)³¹, a renowned market research and consultancy firm specializing in the offshore wind industry, known for its extensive insights into up to 1800 offshore wind projects. The dataset obtained from “4C Offshore” reports a total of 821, 799, and 184 wind farms in Asia, Europe, and other regions, respectively, encompassing both commissioned and under-development projects. This dataset includes key parameters such as latitude, longitude, area, and turbine model types. Additionally, for each wind farm, the dataset specifies the maximum planned installed capacity, while the final installed capacity may be lower. Supplementary Table 1 highlights information on representative wind farms, detailing location, area, installed capacity, wind turbine types, capital expenditure (CAPEX), and investment per MW in U.S. dollars. The wind turbines in Supplementary Table 1 typically have 2–8 MW capacity, with reference wind speeds varying between 42.5 m s⁻¹ and 70 m s⁻¹, depending on the local extreme wind conditions and investment costs.

These wind turbines are manufactured by Siemens, Vestas, Goldwind, and Mingyang Electric, among others. CAPEX ranges from around US\$ 10 million to over US\$ 1 billion, with an investment per MW of US\$ 0.12–0.4 million, aligning with surveys in Angel (2017)⁵⁸.

Methodology for annual maximum wind speed correction

Reanalysis products such as ERA5 underestimate strong winds due to spatial and temporal averaging effects in the numerical model. These effects lead to reduced spectral energy at high frequency ranges compared to observations, for example, an 11% underestimation of annual maximum wind speed in a numerical model with a grid resolution of 25 km compared to anemometer measurements²³. Larsén et al.²³ developed a spectral correction (SC) method to correct the underestimation by filling the missing variability in the high-frequency range of model output. As illustrated in the spectral domain in Larsén et al.²³ (also see example from ERA5 in Supplementary Fig. 12a), observed wind speeds follow a slope of $f^{-5/3}$ for frequencies $f > 1 \text{ day}^{-1}$, whereas model-simulated winds exhibit a steeper decay of f^{-3} . The spectral behavior $f^{-5/3}$ at high frequency is also supported by many theoretical arguments^{59,60}. This discrepancy arises from energy removal by the model's dissipation mechanisms, including filtering and damping schemes. Larsén et al.²³ derived this correction using 10-minute observational wind data from Denmark, with observation heights ranging between 10 m and 60 m. The correction is applied within a targeted frequency range $[f_c, f_e]$, where the lower bound $f_c = 0.8 \text{ day}^{-1}$ addresses the underestimated spectral energy occurring at high frequencies ($f > 0.8 \text{ day}^{-1}$), while the upper bound $f_e = 72 \text{ day}^{-1}$ corresponds to the Nyquist frequency of 10-min temporal resolution. Within this frequency range, the wind spectrum $S(f)$ of reanalysis wind is corrected to follow the observed wind spectrum scaling of $S(f) = f^{-5/3}$. Frequencies outside this range ($f < f_c$ or $f > f_e$) remain unaltered. This approach effectively downscales 1-hour ERA5 wind speed data to a 10-minute equivalent resolution, conforming to the IEC standard, which is the recommended temporal resolution for U_{50} assessment.

By modeling extreme wind exceedances as a Poisson process, the annual maximum wind speed $\bar{U}_{\max}$ can be derived from the zero- and second-order spectral moments (m_0 and m_2) of the corrected spectrum:

$$\bar{U}_{\max} = \bar{U} + \sqrt{m_0} \sqrt{2 \ln \left(\frac{1}{2\pi} \sqrt{\frac{m_2}{m_0}} T_0 \right)} \quad (1)$$

where $\bar{U}$ represents the mean wind speed and T_0 is the reference time period (here $T_0 = 1 \text{ year}$). The i -th order spectral moment is defined as:

$$m_i = 2 \int_0^\infty f^i S(f) df \quad (2)$$

where f represents frequency (Hz) and $S(f)$ denotes the wind speed spectrum. For a comprehensive derivation, refer to Larsén et al.²³.

The SC method has been used in different reanalysis products to assess extreme winds globally^{22,41,42}. However, it systematically underestimates certain regions, especially cyclone-prone regions, as shown by U_{50} comparisons between SC-corrected and raw ERA5 data (Supplementary Fig. 13b). This underestimation may stem from stronger cyclone signals in the high-frequency range, where the spectral slope is steeper than $f^{-5/3}$, which deserves further investigation. While Larsén et al.²² proposed a modified spectral correction for tropical cyclones (SC-TC), our evaluation found persistent underestimation even after its application. Furthermore, the SC-TC method relies primarily on empirical approximations derived from the Holland model by Ott (2005)⁶¹, rather than direct validation against observational data in

tropical cyclone regions. Given these limitations, our study employs the standard SC method for annual maximum wind speed derivation. For cases where the corrected U_{50} remains underestimated, the original ERA5 values are retained as conservative estimates.

Methodology for U_{50} assessment

Wind speed in regions abundant with wind resources has been shown to follow a two-parameter Weibull distribution with an exponential tail in the strong wind component, as established by previous studies^{62,63}. Consequently, the cumulative distribution function (CDF) of extreme wind inferred from the Weibull distribution exhibits a double exponential structure with two parameters, α and β , collectively referred to as the Gumbel distribution⁶⁴. The formula for the CDF is as follows:

$$P(U_{\max}; \alpha, \beta) = e^{-e^{-\frac{(U_{\max} - \beta)}{\alpha}}} \quad (3)$$

where $U_{\max}$ is the annual maximum wind speed derived from the SC method. The fifty-year return period wind speed (U_{50}) is the value of $U_{\max}$ where the CDF of the Gumbel distribution equals 0.98. The rearrangement of the CDF formula allows us to derive U_T at the T -year return period based on the following equation:

$$U_T = -\alpha \ln \left[\ln \left(\frac{T}{T-1} \right) \right] + \beta \quad (4)$$

Pryor and Barthelmie (2021)¹⁷ employed four methods to estimate the parameters of the Gumbel distribution, including the maximum likelihood estimation (MLE), method of moments (MoM), parameters extrapolated from the Weibull distribution estimation (Gumbel-Weibull), and a graphical method. Among these, MLE is considered one of the most robust non-parametric iterative approaches^{65,66}, determining parameters by maximizing the likelihood function under the assumed Gumbel distribution⁶⁷. MoM, while less computationally demanding, is less robust as α and β can be derived from algebraic functions without iteration. The graphical method involves a linear fitting between the ordered observations and the corresponding probability densities, offering a simplified method for distribution inspection⁶⁸ and useful for testing whether the sample of annual maxima is drawn from a single Gumbel distribution⁶⁹. These approaches focus on the annual maximum wind speed, requiring data periods longer than 20 years for stable and convincing estimates⁷⁰.

The Gumbel-Weibull method first derives the parameters of the Weibull distribution using all available hourly data and then calculates α and β based on a linking function¹⁷. This method is suitable when long-term wind speed data are lacking, as the Weibull distribution parameters can remain relatively stable even with short periods (<10 years). However, results from the Gumbel-Weibull method may show discrepancies, especially in tropical regions¹⁷. In light of these considerations, the MLE method is the preferred choice due to its robustness and reliability, even though it involves higher computational demand.

Trend analysis of extreme wind speed

In the context of escalating extreme wind conditions in coastal and oceanic regions, exploring the changes in U_{50} is imperative, as this directly impacts the seamless operation of existing offshore wind farms and the strategic planning of new installations. Leveraging the comprehensive ERA5 dataset, which spans 84 years (1940–2023), we have a valuable opportunity to derive robust estimations of U_{50} under various climate conditions. For estimating U_{50} , previous studies recommend datasets of at least 20 years to ensure sufficient sampling of extreme events. While a 50-year period would theoretically align perfectly with the return

period definition, this duration may mask temporal variability and provide fewer samples for trend detection. Finally, U_{50} is calculated using overlapping 30-year windows shifted annually. To elaborate, the initial U_{50} value is based on data from 1940 to 1969, and the last U_{50} value spans from 1994 to 2023, resulting in a total of 55 derived U_{50} values. The trend analysis of U_{50} is conducted through linear regression applied to these derived values. To validate that the significant changes in U_{50} are not contingent on the specific duration of the dataset, we also conduct trend analyses for U_{50} derived from continuous 20, 30, and 50-year datasets (Supplementary Fig. 4).

Data availability

The hourly mean wind speed, 3-second gust winds, and sea surface temperature are available at <https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-single-levels?tab=form>. The best track data are available at <https://www.ncei.noaa.gov/products/international-best-track-archive>. The data supporting the findings of this study, including the locations of wind farms and the generated U_{50} product and its trend, are publicly available in the Figshare repository at <https://doi.org/10.6084/m9.figshare.29380535>⁷¹.

Code availability

The analysis was conducted using MATLAB R2022b. Analysis scripts, including calculating the U_{50} and its trend, are available at <https://doi.org/10.6084/m9.figshare.29380535>⁷¹.

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Author contributions

Z.Z. conceptualized this study and provided supervision. Methodologies were developed by Z.Z. and Y.Z. The investigation was carried out by Y.Z., Z.Z., and Y.T. Y.Z. drafted the original manuscript. Y.T., J.Y., and Y.C. discussed the results and implications for the manuscript. All authors contributed to the revision of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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